Estimating Aboveground Biomass of Alpine Grassland During the Wilting Period Using In Situ Hyperspectral, Sentinel-2, and Sentinel-1 Data

Rui Guo, Jinlong Gao, Shuai Fu, Yangjing Xiu, Shuhui Zhang, Xiaodong Huang, Qisheng Feng, and Tiangang Liang

***Abstract*— Accurately estimating of grassland above-ground biomass (AGB) during the wilting period is vital in the dynamic monitoring of vegetation survey, carbon storage research, and grazing livestock supplementation. However, previous studies on grassland AGB during the wilting period have rarely involved the integration of ground-based in situ hyperspectral data and satellite images. In this study, we proposed a multisource remote sensing monitoring approach for grassland AGB based on the differential fusion of satellite–ground spectral data from 139 sample sites collected during the grassland's wilting period (September–November) on the northeastern Tibetan Plateau. First, the in situ hyperspectral data and Sentinel-2 images were differentiated fusion by using the nonnegative matrix factorization (NMF) method. Then, the Sentinel-1 synthetic aperture radar (SAR) images were further integrated to develop the random forest (RF) model for estimating AGB in the grassland's wilting period. The results showed that: 1) the NMF-based differentiated fusion model (R^2 = 0.60 and root-mean-square error (RMSE) = 586.56 kg/ha) effectively improved the estimation accuracy of AGB for the grassland wilting period compared with the Sentinel-2 satellite model (R^2 = 0.54 and RMSE = 627.53 kg/ha); 2) the vegetation indices (VIs) derived from short-wave infrared (SWIR) bands are sensitive to variations of grassland AGB during wilting, which have great potential in the estimation of grassland AGB; and 3) the grassland AGB model's performance is only slightly improved by adding Sentinel-1 SAR data and no more significantly positive synergistic effect on the model performance was observed. Overall, this study's proposed satellite–ground collaborative monitoring method integrates the advantages of multisource remote sensing data and is expected to further improve the large-scale and high-accuracy monitoring capability for alpine grassland AGB during the wilting period.**

***Index Terms*— Above-ground biomass (AGB), alpine grassland, data fusion, multisource remote sensing, wilting period.**

I. Introduction

THE Tibetan Plateau's alpine grasslands carry essential ecological, economic, and social values as the primary means of production on which herders depend and the material basis of the animal husbandry economy in pastoral areas [1], [2]. Alpine grasslands comprise hardy herbaceous plants as part of a typical grassland ecosystem. The wilting period is a crucial phenological stage in the life cycle of alpine grassland vegetation. Dynamic monitoring of vegetation above-ground biomass (AGB) in alpine grasslands during this period is vital for efficiently managing grassland resources and optimizing grazing systems.

The most common grazing management on the Tibetan Plateau grassland is nomadic herding, where livestock moves seasonally between summer pastures and winter pastures, and winter grazing is popular [3]. In winter, after the alpine grassland enters the wilting period, the forage gradually dries, the nutrient content decreases, and the livestock feeding pressure increases. The dynamic monitoring of grassland AGB during this period is crucial for herders to reasonably replenish feeds to improve the production efficiency of livestock and the economic benefits of herders and is also conducive to grassland management departments and herders in the scientific management and efficient utilization of grassland resources. The sustainable utilization of wilting grass resources provides economic value while serving as a foundation for restoring grassland ecosystems and preparing the grassland for the return of green in the coming year [4]. In addition, grassland, which comprises a significant portion (31%–43%) of the Earth's surface, is one of the major terrestrial ecosystems except forests, tundra, and agricultural ecosystems and is known for the substantial carbon storage capacity [5]. Accordingly, the dynamic monitoring of grassland AGB is one of the key issues in terrestrial carbon cycle research. During the wilting period, senescent grass leaves have a poor carbon absorption capacity, leading to a reduced amount of sequestrated carbon [6]. To reduce the uncertainty of this

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terrestrial carbon sink, an accurate assessment of the grassland AGB during the wilting period is increasingly necessary [7]. Quantifying vegetation during the wilting period can monitor the grassland's carbon cycling process, an essential component in the vegetation survey and carbon stock study of grassland ecosystems [8]. Therefore, the accurate and real-time acquisition of the spatial dynamics of alpine grassland vegetation during the wilting period is significant for grassland productivity evaluation, ecological environment protection and restoration, and carbon cycle research in grassland ecosystems.

Traditional methods for monitoring grassland AGB—such as destructive harvesting—are time-consuming and energy-intensive [9]. Thus, understanding the large-scale spatial and temporal distribution and variation of the AGB in grasslands is challenging [10], and remote sensing technology provides an important method to monitor grassland AGB dynamically and efficiently [11]. Currently, methods used to estimate grassland AGB include parametric and nonparametric regression. In the primary stage of remote sensing monitoring of the AGB in grasslands, the inversion of AGB at a regional scale is mainly achieved by establishing a linear parametric model between vegetation indices (VIs) and actual measured values of the AGB in grasslands [4], [13]. Over the past two decades, research using machine learning algorithms and remote sensing data for grassland AGB estimation has steadily increased, gradually replacing traditional parametric models as the mainstream method for grassland AGB estimation [14]. Machine learning algorithms, including random forest (RF), decision tree (DT), and gradient boosting (GB) algorithms, have been commonly applied to the construction of the estimation models of the AGB in grasslands [15], [16]. Nevertheless, studies on AGB estimation during the grassland wilting period are relatively few and rely mainly on parametric models. For instance, some have combined remote sensing data to construct linear or exponential models for AGB estimation during the wilting period in northeastern China's western grasslands [4] and inner Mongolia's semiarid grasslands [17]. In addition, previous studies have shown that machine learning algorithms are more effective than linear models for AGB estimation in grasslands [18], [19], [20]. Therefore, analyzing nonparametric models for AGB estimation in grassland ecosystems during the wilting period will become a primary focus of future research.

During the wilting period, both photosynthetic vegetation (PV) and non-PV (NPV) exist in alpine grassland. Due to its high chlorophyll and water content, the canopy spectrum of PV has distinct peaks and valleys in the visible (VIS) and near-infrared (NIR) plateau in the NIR regions, clearly distinguishable from bare soil [21]. However, the reflectance of NPV is the same as that of bare soil in the VIS and NIR regions (350–1400 nm), which makes it challenging to accurately conduct remote sensing AGB estimation during the grassland's wilting period [22]. One study estimated the AGB of PV and NPV in Brazilian Amazonian rangelands using hyperspectral data and showed that lignocellulosic uptake characteristics were highly correlated with the AGB of NPV [8]. This is because NPV's spectral reflectance formed two peaks in the short-wave infrared (SWIR) band (1730 and 2100 nm) due to the absorption of nonpigmented organic molecules [4], [23], [24], which is related to the components, such as cellulose and lignin, that are not found in popular soil [17], [25]. Analytical Spectral Devices, Inc. (ASD) hyperspectral remote sensing data with massive and continuous narrowbands (350–2500 nm) have been demonstrated to provide rich spectral information that can effectively distinguish bare soil from NPV and achieve high-accuracy estimates of grassland AGB during the wilting period [26]. However, due to the study's small observation field of view, it faced in difficulties representing the actual condition of natural grasslands with substantial spatial heterogeneity [27]. Moreover, some uncertainties remained in transposing monitoring results to regional scales due to the limitations of different monitoring environments (differences in field spectral data measurements) and monitoring regions (different types of grasslands).

Most previous remote sensing studies on grassland AGB during wilting were based on satellite images (e.g., MODIS, Landsat TM, and Sentinel-2) [4], [28], [29]. A study on the AGB estimation of NPV in Canadian semiarid mixed grasslands found that VIs constructed based on Sentinel-2 data were generally more closely correlated with the AGB of NPV than those produced using Landsat 8 [30]. By comparing different multispectral sensors for mapping of PV and NPV components in grasslands, the Sentinel-2 Multispectral Instrument (MSI) was shown to have a higher inversion accuracy than Landsat 8 Operational Land Imager (OLI) and GF1 Wide-Field View (WFV) sensors. This is because MSI has higher spatial resolution and more spectral bands than the other sensors, which gives it a significant advantage in extracting PV and NPV components, considerably improving the accuracy of NPV estimation for grassland [23]. The AGB estimation model of withered grass constructed from Sentinel-2 data in western grassland of Northeast China also achieved high accuracy with the correlation coefficient (R) of 0.863 [4]. The Sentinel-2 MSI sensor carries 13 spectral bands, incorporating additional red-edge (RE) and NIR bands that are more suitable for grassland AGB monitoring during the wilting period, in addition to the spectral regions corresponding to Landsat 8 OLI. However, these above multispectral satellites' limited spectral resolution introduces uncertainty to NPV identification and grassland AGB monitoring during the wilting period. The grassland AGB models' estimation accuracy based on hyperspectral bands and their derived VIs generally outperformed AGB estimation models based on multispectral satellites [31], [32]. Thus, fully utilizing of the respective advantages of remote sensing data from multiple (i.e., satellite-based and ground-based) platforms and innovating and developing the synergistic inversion algorithm for satellite–ground multisource remote sensing data are crucial for achieving large-scale, high-precision monitoring of vegetation AGB in alpine grasslands during wilting.

Both ASD hyperspectral data and Sentinel-2 satellite images are optical remote sensing data. In a heavily vegetation canopy condition, optical remote sensing cannot reach the canopy bottom's litter layer due to poor penetration [33], [34]. In open canopy environments, optical remote sensing is affected by litter mass decay at the canopy bottom as well as soil background [22], which creates limitations in the applying of grassland

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vegetation monitoring during wilting. Synthetic aperture radar (SAR) uses active remote sensing by emitting its self-energy and receiving backscattered energy from the ground surface. Compared with optical remote sensing data, SAR has more substantial penetration; is unaffected by clouds, haze, and rain; and is also sensitive to litter mass, crop residues, and NPV [32], [35], [36]. Moreover, ASD hyperspectral provides only biological information about vegetation, while SAR conveys information about the vegetation's physical structure and moisture content of vegetation. NPV typically has a less complex canopy structure than PV, contributing to the difference in backscattering between PV and NPV [32]. In addition, NPV also exhibits lower moisture content, resulting in minimal dielectric capacity and consequently less backscatter [37]. By studying the response of multifrequency and multipolarization radar to crop residues in grasslands or agricultural fields with low canopy, we found that longer wavelength signals tended to penetrate deeper into the soil through the litter layer [35] and were insensitive to litter mass. SAR systems with shorter wavelengths (e.g., X-band and C-band) may be more effective for estimating grassland AGB during wilting [38], but fewer related studies use such systems. Thus, studying the synergy between conventional optical data and SAR data may provide new opportunities for monitoring grassland AGB during the wilting period.

The large-scale monitoring accuracy of grassland AGB during wilting depends heavily on the remote sensing data's spectral, spatial, and temporal resolution [23]. In the design of remote sensing systems, a critical tradeoff occurs between spatial resolution, spectral resolution, and signal-to-noise ratio due to limited incident energy [32], [39]. So far, few satellite platforms provide spectral data with high spatial and spectral resolution, as well as high-frequency temporal sampling simultaneously. However, remote sensing data from different platforms are complementary, and the fusion and synergy of multisource remote sensing data can significantly increase the fine-grained acquisition of surface information from large-scale satellite data [40] and improve the AGB inversion accuracy for regional grasslands during wilting. Accordingly, the choice of the fusion method for the multisource remote sensing data is crucial as it directly affects the accuracy of the fusion model.

In previous studies, researchers have proposed various fusion methods to combine hyperspectral data and satellite images for obtaining high spectral resolution images, high spatial resolution, and high temporal frequency that cannot be directly observed [32], thereby helping to achieve accurate detection, identification, and classification at the regional scale [39], [41], [42], [43], [44]. Among the traditional methods is the component projection gradient method based on the principal component analysis (PCA) and enhanced hue saturation [45], [46]. This method helps use for fusing hyperspectral panchromatic images and low-resolution satellite raster datasets. Since the spectral bands of the multispectral satellite images do not include all the hyperspectral data's bands, spectral distortions are produced during fusion [32]. To improve the spatial resolution of hyperspectral data using multispectral images, Bioucas-Dias et al. [47] proposed the linear spectrum unmixing (LSU) method to solve the problem of common blind source separation by linearly decomposing the remote sensing image into a set of endmember spectra and matching weighted grayscale images. Currently, linear spectral decomposition is mainly based on samples' average reflectance to obtain correction coefficients [48], homogenizing the spectral reflectance differences between samples. This method can produce highly accurate predictions where soil properties are relatively uniform, and reflectance differences between samples are small. However, in ranges with substantial spatial heterogeneity, fusion based on the samples' average reflectance may be insufficient for capturing some critical relationships between samples [49]. Therefore, fusion based on average reflectance is inefficient in simultaneously improving spatial and spectral resolution, somewhat limiting the accuracy of grassland AGB inversion during the wilting period.

Advantageously, the nonnegative matrix factorization (NMF) algorithm offers the advantages of simple implementation [50], pretty interpretation, and intuitive decomposition form and results. It is commonly used for dimensional approximate reduction, feature extraction, and data mining [51], [52]. Compared with linear spectral decomposition methods, the NMF method can capture heterogeneous relationships between different samples to obtain fused images with superior spectral performance and detail retention [53]; they have been widely used in fusion of multisource remote sensing data [40], [48], [54], [55]. For instance, the NMF fusion methods can be readily applied to any combination of spectral datasets with low and high spatial resolution and provide unprecedentedly rich information [56]. Benhalouche et al. [54] found that the fusion spectra based on the NMF method have good spectral and spatial fidelity—significantly better than other methods such as PCA and LSU. Liu et al. [57] applied the NMF algorithm for the first time to isolate soil spectral signals from hyperspectral data and successfully improved the accuracy for soil salinity estimation. Subsequently, the NMF method has been widely used for the differential fusion of ASD hyperspectral data and Sentinel-2 satellite images, vastly improving the soil salinity inversion accuracy using satellite data. Thus, the NMF method shows promise for applications in the fusion of different spectral data; exploring further how to apply this method to the study of grassland AGB estimation is also worthwhile.

Based on multisource remote sensing data for grassland AGB estimation, this study used the observed sample dataset for alpine grassland during the wilting period on the northeastern Tibetan Plateau to further investigate the integration method's applicability and to promote the high accuracy and rapid acquisition of grassland AGB during wilting across a large area. Specifically, the RF algorithm is used to: 1) explorer the potential of the differential fusion algorithm based on NMF to improve satellite data for grassland AGB estimation during the wilting period; 2) investigate the relationship between spectral variables and grassland AGB; and 3) analyze the synergistic effect of Sentinel-1 SAR data on the AGB estimation during grassland's wilting period. This study aims to improve the estimation accuracy of the AGB inversion model for the alpine grasslands. In addition, it seeks to provide the technical

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means for AGB estimation in wilting alpine grasslands and a theoretical basis for grass storage balance, monitoring, and management in the Tibetan Plateau region.

II. Material and Methods

A. Study Area

The alpine grasslands in Menyuan County, Haibei Tibetan Autonomous Prefecture (37°05′N–37°87′N, 100°92′E–102°64′E), and Gannan Tibetan Autonomous Prefecture (100.76°E–103.40°E, 33.11°N–35.57°N) of the northeastern Tibetan Plateau were selected as the experimental regions (Fig. 1). The Menyuan experimental area located in the Alpine Grassland Ecology National Field Scientific Observation and Research Station, Haibei Tibetan Autonomous Prefecture, Qinghai Province. It has a typical plateau continental climate with extended sunshine hours, intense solar radiation, and a vast day–night temperature variation, and other climatic characteristics. It has an average of 2500-h annual sunshine hours, an average yearly temperature of −1.7 °C, and an annual average precipitation of 520 mm. The grassland is mainly of the alpine meadow type; its dominant plant species are *Kobresia myosuroides*, *Carex atrofusca*, and grasses; its primary livestock varieties include sheep and yaks. The Gannan experimental area runs from the northeast to the southwest covering parts of Hezuo, Xiahe, Luqu, and Maqu counties (Fig. 1). The Gannan experimental area is located at the Tibetan Plateau's northeastern edge, with high terrain in the northwest, low terrain in the southeast, and an average altitude of 3000 m above sea level. It also experiences the plateau's continental climate, with concurrent rain and heat and vast vertical differences in heat. It has 1800–2600-h average annual sunshine hours, an average yearly temperature of 1 °C–3 °C, and an average annual precipitation of 400–800 mm. Grassland covers 87% of the land in the experimental area (18 978.3 km^2), which is used mainly for rotational grazing and fencing in all seasons. The dominant species are *Festuca ovina*, *Stipa aliena*, and *Kobresia capillifolia*.

B. Data Collection and Processing

1) Field Data: Seven grassland field surveys were conducted between 2017 and 2022 during the wilting period from September to November in the Tibetan Plateau's northeastern portion. The field survey program used a layout strategy combining fixed and random sample plots; the geographical location of the study area and the layout of sample plots are shown in Fig. 1. Two fixed monitoring sample areas (MY1 and MY2, each with 18 fixed sample plots) were created in the Menyuan experimental area; three fixed monitoring sample areas (GN1–GN3, each with five fixed sample plots) and 62 random sample plots were established in the Gannan experimental area. We collected 94 samples at the fixed sample plots and 62 samples at the random sample plots during the grassland's wilting period. Since some spectral data from Sentinel-2 images were interfered by cloudiness and other factors for the same period, a total of 139 valid samples were obtained for the model development. The sampling method involved creating three 0.5 × 0.5 m sample squares at each sample plot; collecting the above-ground parts of grassland plants in the sample squares; and recording the geographic location, elevation, plant community composition, canopy height, chlorophyll content leaf area index, and fresh weight of each sampling site. The grass samples were brought back to the laboratory, dried at 75 °C for 48 h to a constant weight, and weighed. The AGB (kg/ha) was obtained for each sample plot based on the average of the dry weights of the three sample squares.

2) ASD Hyperspectral Data: The FieldSpec-4 portable hyperspectral ground spectrometer manufactured by ASD was used for vegetation canopy spectrum measurements. Its spectral range is 350–2500 nm with a spectral resolution of 1 nm. The spectral acquisition was performed on dry, windless, and clear days with low cloudiness, and reflectance was timely calibrated using a white spectral panel according to the weather variations of the day. The sensor was placed 1 m directly above the vegetation canopy to be observed with a spectrometer field of view of 25° and an acquisition time of 100 ms between 10:00 and 14:00 when light conditions were favorable. For spectral measurements, the canopy spectral reflectance of vegetation communities was measured ten times using the ASD hyperspectral rotation to reduce the effects of light differences and bidirectional reflectance. We used the Savitzky–Golay filtering algorithm to remove the noise of the measured canopy spectrum from vegetation communities, to improve the signal-to-noise ratio of the spectral signal. Then, the spectral curves were averaged using ViewSpec Pro (version 5.6) software to obtain the actual spectra of each sample square.

3) Sentinel-2 Satellite Data: Launched by the European Space Agency (ESA), the Sentinel-2 satellites are equipped with MSI, which covers 13 bands in the VIS, NIR, and SWIR ranges, and more than three RE bands. Sentinel-2 is composed of 2A satellite (launched in 2015) and 2B satellite (launched in 2017), with high spatial coverage and a revisit cycle of 5 d for the complementary satellites. Data were obtained from the ESA official website (https://scihub.copernicus.eu/dhus/#/home), download Sentinel-2 L2A products covering the study area with low cloudiness for the same period as the ground-based observations, including radiometric calibration, declouding, and atmospheric correction, which we performed using the Sen2cor tool [59]. Then, the spatial resolution of each band was resampled (10 m) by using a nearest neighbor technique.

4) Sentinel-1 SAR Data: The Sentinel-1 consists of two satellites (A and B satellites) and is an SAR sensor carrying C-band. Four imaging modes are currently available, such as interferometric wide swath (IW), wave (WV), extra wide swath (EW), and strip map (SM). This Ground Range Detected (GRD) of IW level-1 products have high spatial resolution (20 m) and radiometric resolution. It contains multivisual intensity data with correlated backward scattering coefficients, which can be widely used for terrestrial AGB inversion [59], [60]. The study obtained GRD products covering our study area from the Google Earth Engine (GEE) cloud platform for the same period, which has two types of polarization: vertical

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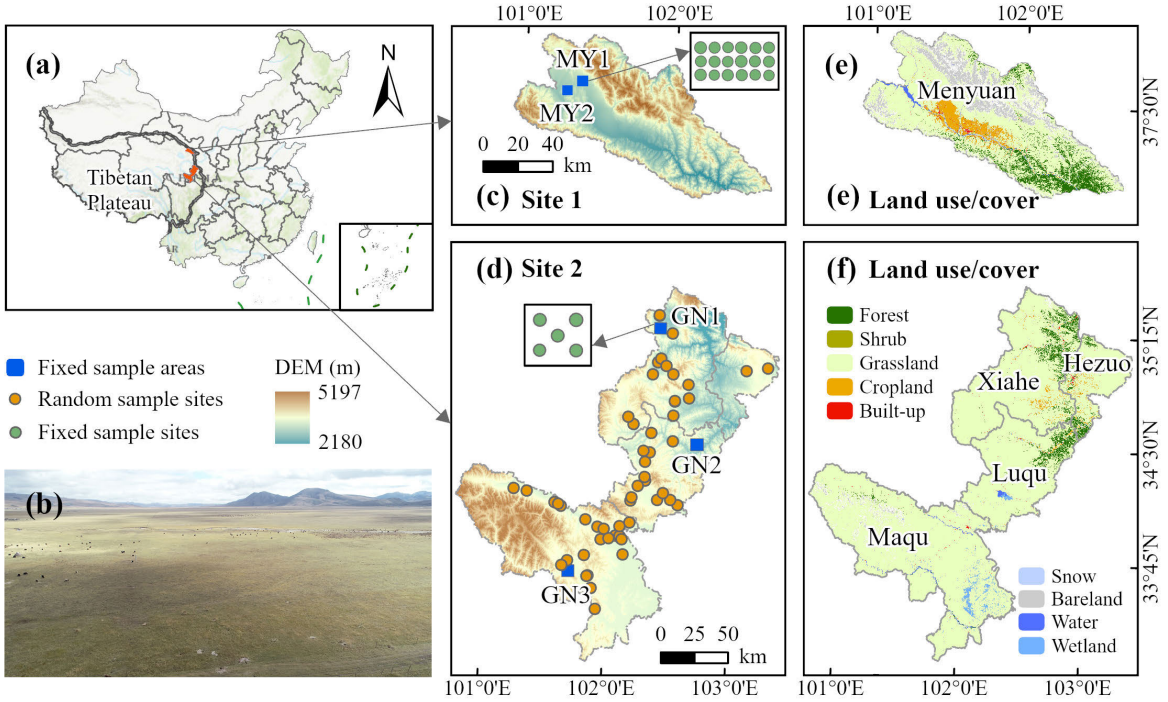


Fig. 1. Study areas. (a) Geographical location of the sites; (b) landscape photograph of alpine grassland during the wilting period; (c) and (d) sample distribution and elevation gradient in sites 1 and 2, respectively; and (e) and (f) land use/cover types in sites 1 and 2, respectively [58].

transmitted vertical received (VV) and vertical transmitted horizontal received (VH). In GEE, the Sentinel-1 images were preprocessed into an analyzable data format using the S1 toolbox (ESA, 2020) with border and thermal noise removal as well as radiometric and terrain calibration [61]. Finally, the cross-polarized backscatter VH and the angle of incidence were selected as feature variables for the AGB during the grass dieback period.

C. Satellite–Ground Remote Sensing Data Fusion

1) Resampling of ASD Hyperspectral to Sentinel-2 MSI: We performed a convolution process on the ASD hyperspectral data of the collected grassland vegetation canopy using the Sentinel-2 spectral response function (SRF) raw files in ENVI (version 5.3) software [62] to generate simulated spectral bands matching the 12 bands of the Sentinel-2 images. Based on the simulated spectra bands and Sentinel-2 MSI bands, we constructed VIs sensitive to grassland AGB during the wilting period according to the equations shown in Table I. Finally, we determined the simulated spectral variables (SSVs) and homologous Sentinel-2 MSI spectral variables (MSISVs) for AGB during the wilting period.

2) Differential Fusion Based on NMF: Lee and Seung [50] proposed the NMF algorithm with in a 1999 Nature article. It has since become a popular data processing approach. As a novel pattern-recognition factor-extraction method, NMF can generate two nonnegative matrices whose multiplication can approximate the original matrix, achieving nonlinear dimensional approximate reduction while keeping the information as constant as possible [75]. Therefore, the NMF method applies to the differential fusion of SSVs and MSISVs during the wilting period of grassland. For a nonnegative spectral matrix V ($m \times n$), we can find the nonnegative matrices W and H so that they satisfy the following equation:

$$V_{(m*n)} \approx W_{(m*r)} H_{(r*n)} \tag{1}$$

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}_i^2}{\text{MSE}}\right) \tag{2}$$

$$\text{MSE} = \frac{1}{mn}\sum_{i=1}^{mn}\left(V_{(m*n)} - W_{(m*r)} H_{(r*n)}\right)^2 \tag{3}$$

where m is the number of features, n is the number of samples, r is the factorization factor number, i is the number of the main factor, and MAX_i is the maximum spectral input value of the initial matrix spectrum. In general, the higher the peak signal-to-noise ratio (PSNR), the smaller the image distortion.

In NMF, setting of the factorization factor number (r) is crucial because it determines the volume of data and the approach level of the initial data. PSNR, an essential indicator of image compression and reconstruction quality [76], [77], is used to determine r [see (2) and (3)]. In this study, the optimal value of r is 4. In the initialization stage of NMF, the multiplicative updating algorithm, which is highly sensitive to the initial value, is selected to decompose the original matrix, and the times of iteration is set to 100 to obtain the initial signature matrix W_0 and the initial sample abundance matrix H_0. In the optimization stage, the alternative least squares algorithm, which uses projected gradients (*cjlin*) and converges faster [78], is used to alternatively decompose the initial matrices W_0 and H_0 to obtain the best signature matrix W and sample abundance matrix H. SSVs contain rich spectral information and better represents the spectral

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TABLE I
SPECTRAL VARIABLES USED FOR ESTIMATING GRASSLAND AGB DURING THE WILTING PERIOD

Spectral variables	Abbreviations	Calculation formulas	References
Normalized difference index-5	NDI5	$(B8A-B11)/(B8A+B11)$	[63]
Normalized difference index-7	NDI7	$(B8-B12)/(B8+B12)$	[63]
Normalized difference tillage index	NDTI	$(B11-B12)/(B11+B12)$	[63]
Soil-adjusted normalized difference tillage index	SNDTI	$(1+L)\times(B11-B12)/(B11+B12+L), L=0.6$	[17]
Normalized difference infrared index	NDII	$(B8-B11)/(B8+B11)$	[64]
Normalized difference senescent vegetation index	NDSVI	$(B11-B4)/(B11+B4)$	[65]
Soil adjusted total vegetation index	SATVI	$(1+L)\times(B11-B4)/(B11+B4+L), L=1$	[17]
Normalized canopy index	NCI	$(B11-B3)/(B11+B3)$	[66]
Soil tillage index	STI	$B11/B12$	[67]
Dead fuel index	DFI	$100\times(1-B12/B11)\times(1-B4/B8A)$	[32]
Green-red index	GRI	$(B3-B4)/(B3+B4)$	[68], [69]
Normalized difference vegetation index	NDVI	$(B8-B4)/(B8+B4)$	[70]
Red-edge NDVI	NDVIred-edge	$(B5-B6)/(B5+B6)$	[71]
Modified red-edge NDVI	mNDVIred-edge	$(B5-B6)/(B5+B6)-2\times B2$	[72], [73]
Angle at near infrared reflectance of IRS	ANIRIRS	$arccos(a^2+b^2-c^2)/(2\times a\times b)$	[74]
Angle at near infrared reflectance of CCD	ANIRCCD	$arccos(d^2+e^2-c^2)/(2\times d\times e)$	[74]

Note: L4; L8; L8A and L11 represent the wavelength of B4, B8, B8A and B11 respectively; $a=\sqrt{(B8-B4)^2+(L8-L4)^2}$; $b=\sqrt{(B8-B11)^2+(L8-L11)^2}$; $c=\sqrt{(B11-B4)^2+(L11-L4)^2}$; $d=\sqrt{(B8A-B4)^2+(L8A-L4)^2}$; $e=\sqrt{(B8A-B11)^2+(L8A-L11)^2}$

characteristics of grassland AGB during the wilting period, and MSISVs have the significant advantage in terms of high spatial resolution. Thus, the signature matrix W_{SSV} of SSVs was used as the base spectral matrix for differential fusion; the sample abundance matrix H_{MSISV} of MSISVs was used as the differentiated coefficients of the fused samples, which were multiplied to obtain the differential fusion of spectral variables (DFSVs). The relevant calculations were performed in MATLAB R2022a.

D. Construction and Accuracy Evaluation of the Inversion Model of Grassland AGB in the Wilting Period

1) Variables Selection: This study used the least absolute shrinkage and selection operator (LASSO) method to select the 22 feature variables obtained from the original bands and VIs to reduce the data dimension, reduce redundant data, and improve computational efficiency. Tibshirani [79] originally proposed LASSO as an effective method to integrate variable combination and model selection, which can achieve variable selection through "sparsity" and "model simplification." This method gets a well-developed model by building a penalty function that allows the coefficients to be compressed to zero without significant effects on variable selection and estimation. LASSO's advantages in selecting variables include its ability to effectively reduce the number of variables, improve the computational efficiency, and reduce the complexity of the model, thus avoiding overfitting [80]. This study used the LASSO program in R Studio to select relevant variables for grassland AGB during the wilting period.

2) Inversion Model Based on RF Algorithm: Breiman [81] originally proposed the RF model as a supervised learning algorithm for solving nonlinear problems. The method employs the bootstrap sampling method to establish a DT. After many DTs are established, voting is used to measure the final prediction. Due to its strong resistance to overfitting, the RF algorithm is widely used in remote sensing studies of grassland AGB estimation [82], [83]. This study used the TreeBagger regressor in MATLAB (R2022a) to construct the RF model. Before selecting the parameter value, we used the grid parameter search method to optimize two key parameters in the RF model—NumTrees and MinLeafSize. The NumTrees values were tested from a default setting of 500–10 000 trees with intervals of 500, and we hypothesized that the RF model has N variables and that MinLeafSize is increased from 1 to N in intervals of 1 each time. Ultimately, two key parameters in the RF model—NumTrees and MinLeafSize—were set to

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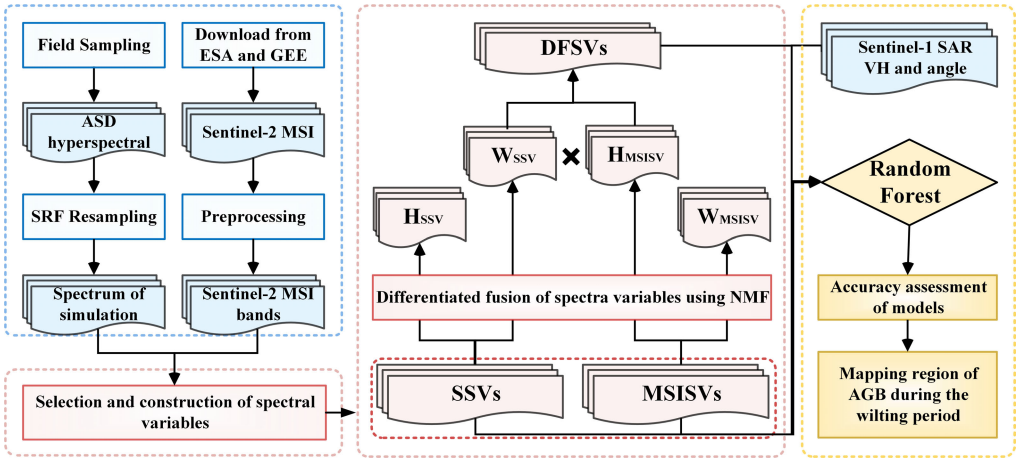


Fig. 2. Methodological flowchart.

1000 and 2, respectively, while the other parameters were set to their default values. The RF algorithm also calculates the importance values of input variables based on the estimation error of the out-of-bag sample. The larger the importance value of the spectral variables, the more critical they were for estimating grassland AGB during wilting.

3) Accuracy Evaluation of Model: This study used a tenfold cross-validation method to validate the models' estimation accuracy. This method's basic principle is to randomly divide the sample dataset into ten groups, and nine groups are used as training datasets, while the remaining group serves as the testing dataset. This process is repeated ten times to ensure that each group is used for both training and testing. The model's performance is evaluated by averaging the results from the ten iterations [84]. The coefficient of determination (R^2), root-mean-square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) were used to evaluate the model. R^2 indicates the degree of correlation between the independent and dependent variables. RMSE is used to evaluate the error between the estimated value and the measured value. MAE is the average of the absolute value of the deviation of all measured values from the arithmetic mean values, which can forestall the problem of errors offsetting each other [85]. Thus, it can accurately reflect the level of the real prediction error. MAPE can reflect the average percentage size of MAE. The calculation formulas of R^2, RMSE, MAE, and MAPE are given in the following equations:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - y_i)^2}{n}} \tag{5}$$

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n}|x_i - y_i|}{n} \tag{6}$$

$$\mathrm{MAPE} = \frac{\sum_{i=1}^{n}\left(\frac{|x_i - y_i|}{y_i}\right) \times 100\%}{n} \tag{7}$$

where x_i and y_i are the AGB estimated values and AGB measured values, respectively, $\bar{y}$ is the mean of the AGB measured values, and n is the sample volume. The larger R^2, the higher the confidence of the model estimation, and the smaller the RMSE, MAE, and MAPE, the better the model fit.

4) Methodological Flowchart: The method of remote sensing estimation of grassland AGB during the wilting period is shown in Fig. 2. First, the ASD hyperspectral data were resampled into simulated spectra based on SRF. Based on the simulated spectral bands and Sentinel-2 bands, we constructed VIs and selected feature variables, respectively, to obtain the SSVs and satellite spectral variable MSISVs. Then, the NMF of SSVs and MSISVs was performed to obtain the DFSVs. We constructed the RF model using the four types of spectral factor groups of SSVs, MSISVs, DFSVs, and DFSVs_S1 (DFSVs + VH and angle bands of Sentinel-1) and evaluated its accuracy. We applied the optimal model to invert the spatial distribution pattern of grassland AGB during the wilting period in the study area.

III. Results

A. Variations in AGB and Reflectance Spectra Among Different Wilting Stages

Statistical analysis of the AGB during wilting of alpine grassland in the study area is shown in Fig. 3. The alpine grassland AGB in the study area had a maximum value of 5237.07 kg/ha in the early wilting stage (September) and a minimum value of 632.13 kg/ha in the late wilting stage (November), with standard deviations ranging from 543.04 to 831.97 kg/ha. In terms of time, the alpine grassland began wilting in September, with a general predominance of PV and the mean value of AGB reaching a maximum of 3041.72 kg/ha. In October, the grassland entered the wilting stage's middle portion, the productivity of the grassland decreased, and the mean value of AGB decreased compared to September. In November, the grassland entered the late wilting stage when some vegetation became litter to be decomposed. During this stage, the grazing utilization of the grassland continued, and the mean value of AGB had a minimum value of 1615.07 kg/ha. In general, the mean value of AGB of alpine grasslands in the study area continued to decrease uniformly

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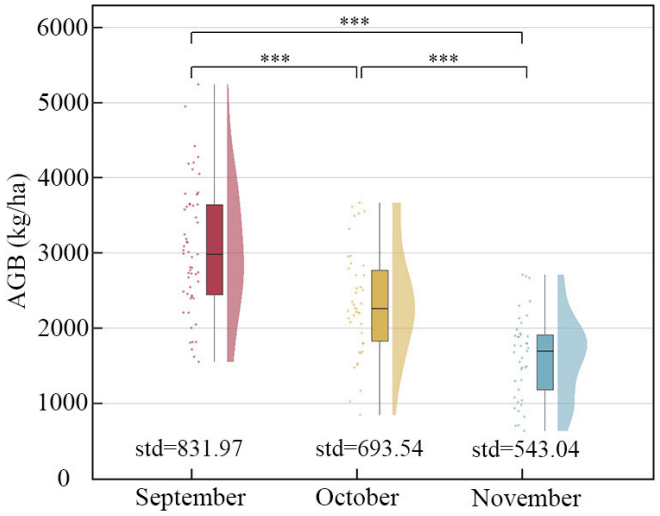


Fig. 3. Variations in grassland AGB among different wilting stages. $***$: $p<0.001$.

after entering the wilting period, and the maximum, minimum, and standard deviations showed the same trend.

The hyperspectral reflectance curves of vegetation of alpine grassland during the wilting period was shown in Fig. 4. We found that the spectral characteristics of the grassland differed at various wilting stages. Generally, from the early wilting stage (September) to the late wilting stage (November) of alpine grassland, the cover and proportion of NPV gradually increased, the chlorophyll and water content of vegetation gradually decreased, the lignin and cellulose content grew, and the spectral reflectance characteristics of vegetation transformed from PV to NPV. In the early wilting stage (September), PV dominated the alpine grassland with high chlorophyll (mainly chlorophyll a and chlorophyll b) content in the leaves, and two absorption valleys (blue absorption valley and red absorption valley) and one green reflection peak appeared in the VIS band of 350–680 nm. In the RE band of 680–760 nm, the reflectance rose rapidly. The NIR band of 780–1160 nm had an NIR plateau with a maximum reflectance of 0.42. In the SWIR bands (1460–1750 and 1980–2300 nm), and PV has lower reflectance values due to its higher water content, which blurs the absorption properties of cellulose and lignin. In the middle of the wilting period (October), the grassland already exhibited extensive wilting; accordingly, the spectral absorption characteristics in the VIS and NIR regions were weakened. Two reflectance peaks emerged, and the reflectance values were higher than in September in the SWIR region due to the absorption characteristics of cellulose and lignin. In the late wilting season (November), the alpine grassland was entirely wilted. The peaks and valleys and the NIR plateau wholly disappeared, and the reflectance values showed a constantly rising trend in the VIS and NIR regions due to the lack of absorption of pigment molecules. In the SWIR (1663 and 2201 nm) region, cellulose and lignin absorption were apparent, with maximum reflectance values (0.33 and 0.23, respectively) at the two reflectance peaks.

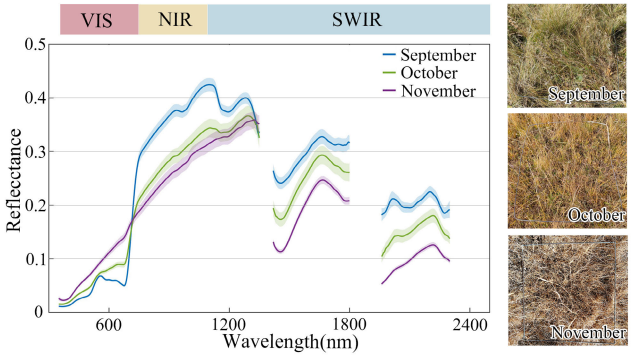


Fig. 4. Spectral reflectance of grassland vegetation during different wilting stages. VIS: visible region. NIR: near-infrared region. SWIR: short-wave infrared region.

B. Spectral Variables for Grassland Biomass During the Withered Period

We used LASSO to select $DFSV_S$, retaining the spectral information that was relatively valid for the AGB estimation in the wilting period. The optimal combination of variables retained 16 spectral variables, including seven spectral bands and nine VIs. The spectral bands included the VIS (B2 and B3), NIR (B5–B7), RE (B8A), and SWIR (B11) bands. The VIs included normalized difference tillage index (NDTI), soil-adjusted NDTI (SNDTI), normalized difference senescence vegetation index (NDSVI), soil-adjusted total vegetation index (SATVI), normalized canopy index (NCI), soil tillage index (STI), dead fuel index (DFI), normalized difference vegetation index (NDVI), and angle at near-infrared reflectance of IRS (ANIRIRS). We performed a correlation analysis of the bands and VIs of the optimal combination of variables to obtain the coefficient of determination R^2 for each spectral variable with wilting-period grassland AGB (Fig. 5). We found that most variables had high R^2, with the highest R^2 of 0.4 for B2 and B11, which correlated best with wilting-period AGB. The advantage of using LASSO to improve model performance was further confirmed as it eliminated redundant variables while retaining the maximum possible spectral information.

C. Accuracy Evaluation for Different Models

According to the spectral variables selected using LASSO, we constructed RF models for AGB estimation for wilting-period alpine grassland based on the three types of spectral factor groups (SSVs, MSISVs, and DFSVs). The models' accuracy was evaluated by adopting tenfold cross validation, with the results shown in Table II. The ASD model based on SSVs had the highest accuracy with an R^2 of 0.64 and an RMSE of 558.97 kg/ha. The Sentinel-2 model based on MSISVs had the lowest accuracy, with an R^2 of 0.54. Compared to the Sentinel-2 model, the accuracy of the differential fusion model based on DFSVs had improved significantly, with R^2 increasing by 11.11% from 0.54 to 0.60 and RMSE, MAE, and MAPE decreasing by 6.53%, 5.87%, and 5.89%, respectively. This finding illustrates the excellent performance of the differential fusion model in

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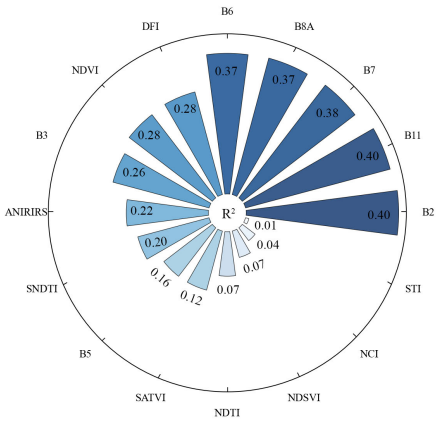


Fig. 5. Coefficient of determination between spectral variables and grassland AGB during the wilting period.

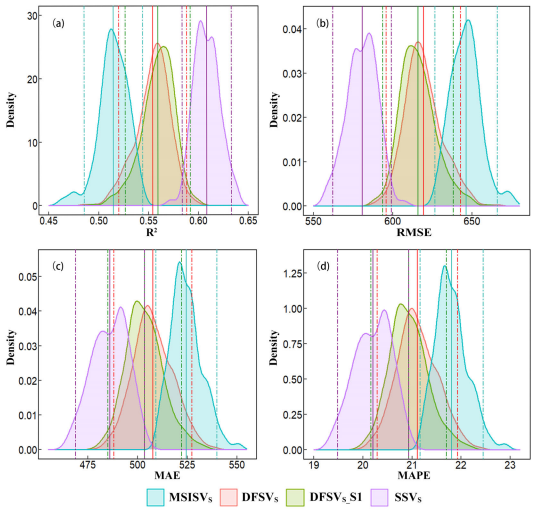


Fig. 6. Density distributions of (a) R^2, (b) RMSE, (c) MAE, and (d) MAPE for the grassland AGB estimation models during the wilting period. The dashed line indicates the mean value. MSISVs: Sentinel-2 models. DFSVs: differential fusion models. DFSVs_S1: synergistic models. SSVs: ASD models.

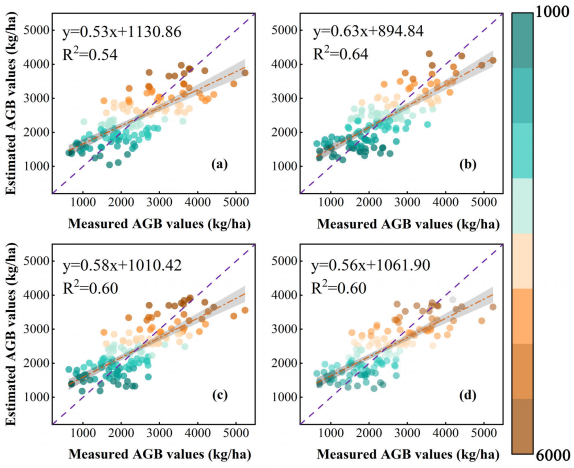


Fig. 7. Scatter diagrams of models. (a) Sentinel-2 models. (b) ASD models. (c) Differential fusion model. (d) Synergistic model.

wilting-period grassland AGB estimation. In addition, the VH and angle of incidence of Sentinel-1 SAR correlated well with grassland AGB [77]. Therefore, we constructed the synergistic model of Sentinel-1 and DFV_S by adding the VH and angle bands of Sentinel-1 to the DFV_S to investigate Sentinel-1 SAR's synergistic effect in wilting-period grassland AGB estimation.

We constructed the AGB estimation models based on the four spectral factor groups for grassland's wilting period. We repeated this 200 times to obtain the density distribution for the models' accuracy evaluation (Fig. 6). This process revealed that the ASD models had the highest accuracy (the mean value of R^2 was 0.61); the Sentinel-2 models had the lowest accuracy (the mean value of R^2 was 0.52). The differential fusion and synergetic models improved the accuracy of the Sentinel-2 models with mean R^2 values of 0.55 and 0.56, respectively, consistent with the optimal model's accuracy evaluation results (Table II). The synergistic models had better robustness (higher R^2 and lower MSE, MAE, and MAPE) compared to the differential fusion models. In the differential fusion models and the synergistic models, the mean values of R^2 were between 0.55 and 0.56, and the mean values of MSE, MAE, and MAPE ranged from 615.92 to 619.40 kg/ha, 503.54 to 507.78 kg/ha, and 20.93% to 21.11%, respectively. Broadly, the two types of models were similar in accuracy. Both models better reflected the spatial and temporal distribution of grassland AGB during the wilting period.

Scatter plots of estimated and measured values were used to evaluate the models' prediction ability for model's accuracy validation; the results are shown in Fig. 7. Among the four types of models, the Sentinel-2 model had the worst accuracy ($R^2 = 0.54$), and the ASD model had the best accuracy ($R^2 = 0.63$), consistent with the machine learning algorithm's evaluation results. The differential fusion and the synergistic models had high estimation accuracy ($R^2 = 0.60$) while achieving spatial upscaling. Both had excellent applicability, with most sample points uniformly distributed on both sides of the 1:1 line and only a few sample points offset. Therefore, upon comprehensively considering the complexity and efficiency of the models, the differential fusion model was finally determined as the optimal model for wilting-period AGB inversion under the condition of similar model accuracy.

D. Spatial Distribution Characteristics of Grassland AGB During the Wilting Period

The spatial distribution of grassland AGB in the study area in November 2022 is shown in Fig. 8. The amount of grassland AGB (kg/ha) during the wilting period varied from low to high as the map color shifted from red to green. The analysis results showed that the minimum value of grassland AGB was 1103.00 kg/ha, and the maximum value was 2976.00 kg/ha in November 2022 in the study area. It was consistent with the observed values of AGB at that time (a minimum value of 632.13 kg/ha and a maximum value of 2708.93 kg/ha), indicating the excellent mapping accuracy of grassland AGB during the wilting period in the study area.

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TABLE II
ACCURACY EVALUATION OF THE GRASSLAND AGB ESTIMATION MODEL DURING THE WILTING PERIOD

Model	R^2	RMSE (kg/ha)	MAE (kg/ha)	MAPE (%)
ASD model	0.64	558.97	467.96	19.45
Sentinel-2 model	0.54	627.53	514.93	21.41
Differential fusion model	0.60	586.56	484.70	20.15
Synergistic model	0.60	586.32	481.43	20.01

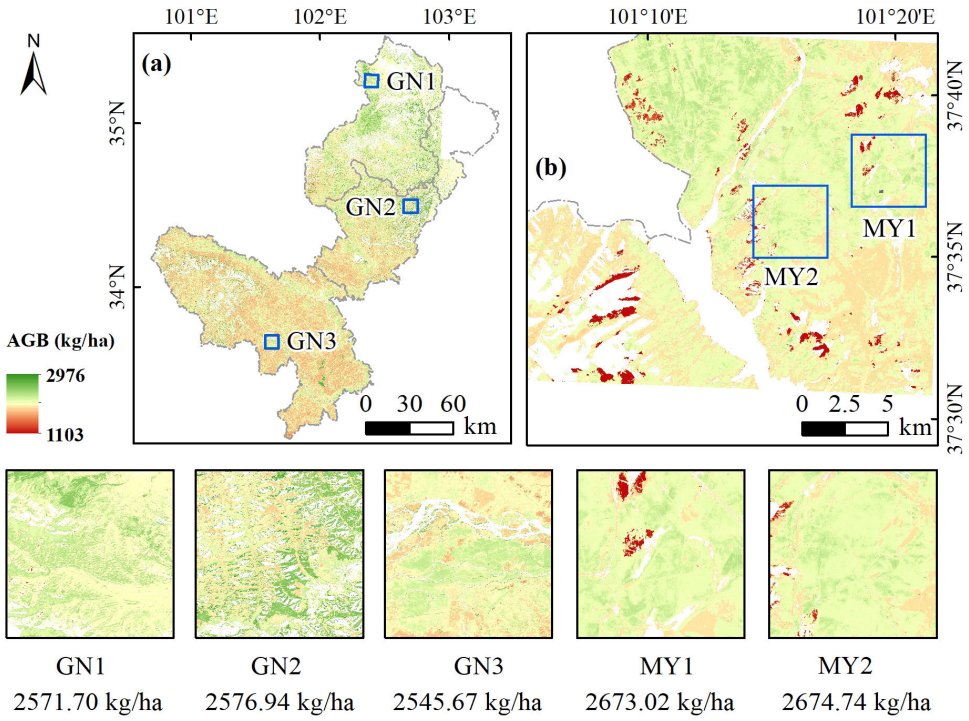


Fig. 8. Spatial distribution characteristics of grassland AGB in (a) Gannan (site 1) and (b) Menyuan (site 2) experimental areas in November 2022. The AGB values represent the mean values of grassland AGB of the corresponding fixed sample plots.

Regarding spatial distribution, significant regional differences were found in grassland AGB in both the Gannan and Menyuan experimental areas. The grassland AGB in the former area decreased gradually from northeast to southwest due to topographic factors; lower values of AGB appeared in Luqu and Maqu counties, which are generally higher in elevation and have more mountains than the other counties. In addition, the AGB values in Xiahe County were typically higher than those of other areas.

IV. DISCUSSION

A. Spectral Indices for Grassland AGB Estimation During the Wilting Period

In remote sensing methods, the difference in grassland AGB monitoring between the growing and wilting season mainly arises due to the difference in spectral reflectance among modeling principles. Most studies have established grassland AGB models in the growing season using remote sensing data because grassland vegetation responds more to spectral bands and VIs, and correlations between AGB and spectral variables are higher in the growing season. Still, these characteristics were difficult to find during the wilting period of grassland [4]. In this study, we selected 16 spectral variables sensitive to grassland AGB during the wilting period and developed the optimal inversion model of wilting-period grassland AGB. To further analyze the connection between the optimal model's performance and the input spectral variables, the contributions to the model of seven spectral bands and nine VIs were calculated separately [Fig. 9(a)]. The B11 band contributed the most to the model, explaining 67% of the grassland AGB during the wilting period. In addition, importance values for the model ranged from high to low: B2, B3, B5, SATVI, B6, STI, SNDTI, DFI, B8A, B7, NCI, NDVI, ANIRIRS, NDTI, and NDSVI, all of which had values of more than 0.3 and had ample potential for wilting-period grassland AGB inversion.

Upon comparing the spectral variables of the optimal model, the SWIR indices showed a consistent and relatively favorable performance in quantifying grassland AGB during the wilting period [Fig. 9(a)]. This study constructed the NDTI (calculated by B11 and B12), SNDTI (calculated by B11 and B12), STI

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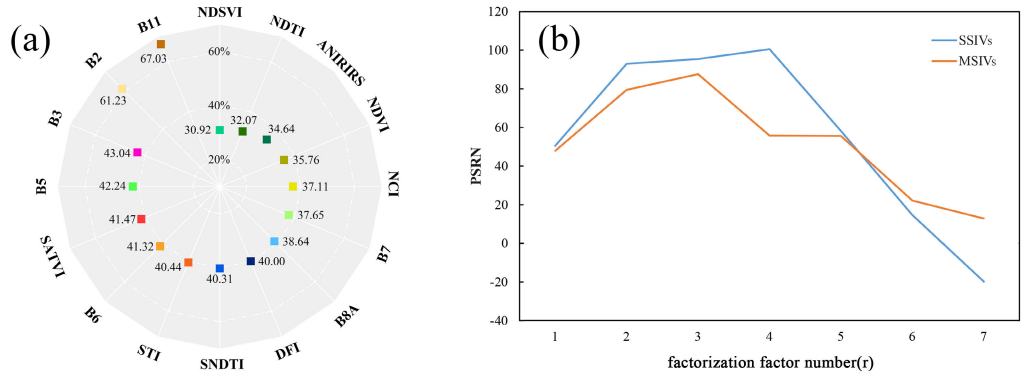


Fig. 9. (a) Contributions of variables to the optimal model. (b) Variations in the PSNR of SSIVs and MSIVs with factorization factor number (r).

(calculated by B11 and B12), NDSVI (calculated by B11 and B4), SATVI (calculated by B11 and B4), NCI (calculated by B11 and B3), DFI (calculated by B12, B11, B8A, and B4), and ANIRIRS (calculated by B11, B8, and B4) from SWIR bands or the combination of SWIR and other bands in the inversion model. Accordingly, the SWIR indices significantly contributed to grassland AGB estimation during wilting. Along similar lines, a quantification study of withered material in arid grasslands' nongrowing period found that NDTI, SNDTI, and NDSVI based on SWIR bands showed better prediction performance than other indices, confirming our conclusion [17]. Unsurprisingly, the B11 band in the SWIR region served as the best spectral band for quantifying wilting-period grassland AGB and had the highest correlation with grassland AGB (Fig. 6). In addition to the SWIR bands, the NIR (B8 and B8A) bands were also essential variables for constructing the above VIs with high sensitivity to the variations of wilting grassland AGB. In addition, we found that DFI and ANIRIRS containing both SWIR bands and NIR bands were more closely correlated with wilting-period grassland AGB than other SWIR indices. Studies on the AGB of NPV in semiarid mixed grasslands in Canada have shown that differences in the NIR and SWIR bands are pivotal for grassland AGB estimation, aligning with our conclusions [30].

The VIS (B2 and B3) and the RE (B5–B7) bands significantly contributed to grassland AGB inversion during wilting. Adding the RE band (B5) can substantially reduce the saturation effect in high biomass areas, improving the wilting-period AGB model's estimation accuracy [61]. Some researchers have suggested that the slope of the RE position varies with moisture content and age of the litter mass, which can affect its application in estimating the AGB of NPV [86]. This is inconsistent with the conclusions of this study. However, compared to the quantification of NPV, we included the quantification of both NPV and PV during the grassland's wilting period. Like the VIS bands, the RE bands are more sensitive to variations in chlorophyll content [87], [88], which can identify PV in grasslands during the wilting period and reduce the error of AGB estimation. Accordingly, the VIS and RE bands are significant for quantifying wilting grassland AGB.

The NDVI was sensitive to variations in grassland AGB during the wilting period. In this study, the contribution of NDVI to the AGB estimation model during the wilting period was high, with an importance value of 0.36, indicating that NDVI is equally effective in the AGB estimations during the wilting and growing seasons. Notably, the correlation between NDVI and grassland AGB is associated with the percentage of NPV [32]. NDVI was positively correlated with AGB when the NPV ratio was $<20\%$, but no significant correlation was found when the NPV ratio ranged from 20% to 80%. However, a strong negative correlation was observed between NDVI and AGB when the NPV ratio was $>80\%$. Therefore, in future studies for monitoring AGB, we can choose different VIs according to varying percentages of NPV.

B. NMF-Based Differential Fusion Method

The NMF algorithm has an extensive application in remote sensing data analysis, such as mixed pixel unmixing and multisource remote sensing data fusion, because of its good spectral retention and spatial fidelity [89], [90]. Unfortunately, the literature offers few examples of using the NMF-based spectral fusion method to quantify grassland AGB. In this study, we took the lead in using the NMF method to solve the problem of the high-precision inversion of grassland AGB during the wilting period, which has long confounded efforts in the remote sensing estimations of grassland vegetation parameters. Our results indicate that the NMF-based differential fusion method enriched the satellite spectral information, resulting in a notable improvement in the accuracy of grassland AGB estimation on satellites during the wilting period ($R^2 = 0.60$ and RMSE $= 586.56$ kg/ha) while enhancing the spatial scale. Instead of the Sentinel-2 satellite model, Chen et al. [48] used the NMF-based differential fusion model to invert soil salinity, with a 16% improvement in model accuracy (from 0.62 to 0.72). Similarly, we used the difference fusion model to invert the grassland AGB during the wilting period, and R^2 of the model was improved by 11% (from 0.54 to 0.60). This indicates that the NMF-based differential fusion method not only applies to soil salinity monitoring but also has excellent potential for AGB estimation during

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the wilting period. Accordingly, the NMF provides a novel approach for high-precision fusion of multisource remote sensing data.

NMF is an unsupervised algorithm that works by iteratively obtaining two nonnegative matrices that approximate the initial matrix [91]. Thus, the performance of the NMF-based differential fusion model is usually very sensitive to the selection of parameters, such as the initial matrix, loss function, and factorization factor number [92]. The initial matrices usually comprise the ASD hyperspectral data acquired from field observations and the satellite images downloaded from a satellite platform. We can improve the model's performance by enhancing the accuracy of the original data through preprocessing such as removing abnormal values and noise signals [93]. The factorization factor number is an essential parameter in NMF. We used PSNR to evaluate the results of NMF based on different factorization factor numbers [Fig. 9(b)].

The PSNR of both $SSIV_S$ and MSISVs showed rising and then falling trends with an increase in the factorization factor number. First, the rise in the factorization factor number can better approximate the original data and improve the accuracy and quality of the reconstructed matrix, thus improving the PSNR. As the factorization factor number continued to increase, the PSNR remained essentially constant of Chen et al.'s process of NMF [48] and began to decrease in our study. It is because the hyperspectral data were observed under laboratory conditions, in which the light source, ambient temperature, and humidity were relatively stable, and the data were less noisy. In our study, the hyperspectral data were observed in the field and often included significant noise, and NMF is a method that cannot handle noise [41]. PSRN is more sensitive to noise than other evaluation indicators such as structural similarity index measure (SSIM) [94]. When the factorization factor number is too large, the noise may be decomposed into the matrices as well, resulting in a decrease in the reconstruction matrices' accuracy and a decrease in the PSNR values. Moreover, when the factorization factor number increases, the model's complexity also increases, leading to the appearance of overfitting and a reduction in PSNR values [95]. Consequently, the data can be processed to remove noise before decomposing, and a reasonable factorization factor number can be selected by considering the application scenario, data characteristics, and noise situation to prevent the overfitting of the model.

C. Limitations and Prospects

The study has some limitations regarding remote sensing data selection and research methods. Compared with the differential fusion model, the synergistic model, which added Sentinel-1 SAR, showed only a slight improvement in accuracy and did not fully reflect the synergistic effect of Sentinel-1 SAR. However, Li and Guo [96] used the cross-polarization ratio (VH/VV) from Radarsat-2 imagery to estimate the AGB of NPV and total AGB (R^2 of 0.70 and 0.51, respectively) in semiarid mixed grasslands in Grassland National Park, showing the potential of SAR data for estimating AGB in grassland. It is because the ability of SAR to estimate grassland AGB is affected by various factors, such as wavelength, angle of incidence, polarization mode, surface roughness, and surface moisture content. The angle of incidence range of Radarsat-2 is 49°–60°, while the angle of incidence range of Sentinel-1 SAR is 20°–46°, and that of Sentinel-1 SAR is about 30° near our study area. The lower incidence angle may allow SAR to penetrate through the litter layer to the soil surface, reducing the sensitivity of SAR to grassland AGB during the wilting period. Furthermore, as the angle of incidence and wavelength increase, the SAR data are more strongly influenced by surface roughness in NPV quantification studies [97], [98]. Therefore, investigating whether a higher angle of incidence is beneficial to estimating grassland AGB during the wilting period may also be worthwhile.

Our study's differential fusion method based on NMF shows promise for the application of grassland AGB estimation in the wilting period. However, this method is accompanied by some challenges. The NMF algorithm has a limitation in that it can only be applied to nonnegative spectral variables, which restricts the use of many negative VIs sensitive to AGB during the wilting period. The semi-NMF and convex NMF proposed by Ding et al. [99] relaxed the restriction on the nonnegativity of NMF by allowing the input matrix and the decomposed matrix to have negative values, thereby expanding the potential application of NMF. In addition, the NMF algorithm often requires extensive iterations to solve complex optimization problems, which is computationally intensive and requires a high time cost [32]. Therefore, future research on differential fusion model based on NMF must focus on optimizing the model parameters, relaxing the model's nonnegative restrictions, and improving the model's computational efficiency.

In addition, the pixel resolution of the Sentinel-2 satellite is 10 × 10 m, which is significantly larger than the ASD hyperspectral observation sample (0.5 × 0.5 m). The huge spatial difference inevitably produces errors in the fusion process. Furthermore, the limited number of samples acquired by ASD hyperspectral makes samples not representative enough for the development of an efficient model for the quantification of grasslands AGB during the wilting period in a large study area. Imaging hyperspectral enables the acquisition of a full reflectance spectrum of each pixel in the image, which reduces the spectral confusion between NPV and soil and allows for better biophysical and chemical measurements on complex surfaces such as grazed grasslands. Imaging hyperspectral mainly includes unmanned aerial vehicle (UAV) hyperspectral and satellite-based hyperspectral. Compared with the local point provided by ASD hyperspectral, UAV hyperspectral can obtain higher spatial resolution and higher quality images faster and more conveniently, enriching the sample dataset [100], [101]. At this spatial scale, UAV hyperspectral involves more flexible data acquisition methods, and spectral images of different pixel sizes can be acquired by adjusting the flight altitude [102], to match the spatial resolution of different satellites. Meanwhile, with the advent of new satellite-borne imaging spectrometers [103] allowing to measure the absorption properties of lignocellulose, novel technical means have been provided for the AGB quantify during the grassland wilting period. Among them, Hyperion and Probe-1 have good applications in monitoring such as pasture NPV and crop

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residues [104], [105]. The advisory group of the Copernicus Hyperspectral Imaging Mission for the Environment provided a set of products that prioritized vegetation and soil variables, including lignin and cellulose variables for quantifying NPV in grasslands [106]. Therefore, we predict that image hyperspectral technology will become a prevalent means in future differential fusion of hyperspectral data and satellite multispectral satellite data.

V. Conclusion

Through field observations of alpine grassland AGB during the wilting period, we proposed a differential fusion method of ASD hyperspectral and Sentinel-2 satellite data to regionally assess the estimation accuracy and mapping potential of wilting-period grassland AGB at a regional scale. Our results indicated that the NMF-based differential fusion model is the optimal inversion model ($R^2 = 0.60$ and RMSE $= 586.56$ kg/ha). The differential fusion method improved the inversion accuracy of grassland AGB during the wilting period compared with the Sentinel-2 satellite ($R^2 = 0.54$ and RMSE $= 627.53$ kg/ha). In addition, the VIs derived from SWIR bands performed well in grassland AGB estimation.

Moreover, we analyzed the synergistic effect of Sentinel-1 SAR data on the prediction performance of the estimation model of wilting-period grassland AGB. The results showed that Sentinel-1 SAR data did not significantly and positively enhance the prediction accuracy of wilting-period grassland AGB. Generally, this present study outlines ideas to improve the generalizability of AGB estimation model in grassland's wilting period and the sustainable use and management of alpine grassland resources. Subsequent research may adopt UAV hyperspectral images and different SAR data while considering the incidence angle, wavelength, and polarization to explore the potential for AGB estimation during the wilting period.

References

[1] J. Liu and J. Yang, "Research on maximum yield prediction and evaluation method of alpine meadows in Haibei area of Qinghai Province," *Qinghai Grassland Ind.*, vol. 29, no. 4, pp. 2–8, Nov. 2020.

[2] X. J. Zhao, G. M. Zhang, and G. J. Zhao, "Current situation of grassland resources in Haibei Prefecture and measures for sustainable utilization," *Grassland Animal Husbandry*, no. 3, pp. 55–57, Mar. 2007.

[3] W. Wei et al., "Suitable grazing during the regrowth period promotes plant diversity in winter pastures in the Qinghai–Tibetan Plateau," *Frontiers Ecol. Evol.*, vol. 10, Oct. 2022, Art. no. 991967, doi: 10.3389/fevo.2022.991967.

[4] P. Zang, Y. Zhang, Z. Chen, G. Hou, Z. Liu, and X. Lu, "The inversion modeling and aboveground biomass mapping of withered grass changes in the western grassland of Northeast China," *Frontiers Earth Sci.*, vol. 10, Jan. 2023, doi: 10.3389/feart.2022.1031098.

[5] D. Xu, X. Guo, Z. Li, X. Yang, and H. Yin, "Measuring the dead component of mixed grassland with Landsat imagery," *Remote Sens. Environ.*, vol. 142, pp. 33–43, Feb. 2014, doi: 10.1016/j.rse.2013.11.017.

[6] L. Royimani, O. Mutanga, and T. Dube, "Progress in remote sensing of grass senescence: A review on the challenges and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7714–7723, 2021, doi: 10.1109/JSTARS.2021.3098720.

[7] H. Ren and G. Zhou, "Estimating senesced biomass of desert steppe in inner Mongolia using field spectrometric data," *Agricult. Forest Meteorol.*, vol. 161, pp. 66–71, Aug. 2012, doi: 10.1016/j.agrformet.2012.03.010.

[8] T. G. Morais, R. F. M. Teixeira, M. Figueiredo, and T. Domingos, "The use of machine learning methods to estimate aboveground biomass of grasslands: A review," *Ecol. Indicators*, vol. 130, Nov. 2021, Art. no. 108081, doi: 10.1016/j.ecolind.2021.108081.

[9] M. A. White, G. P. Asner, R. R. Nemani, J. L. Privette, and S. W. Running, "Measuring fractional cover and leaf area index in arid ecosystems: Digital camera, radiation transmittance, and laser altimetry methods," *Remote Sens. Environ.*, vol. 74, no. 1, pp. 45–47, Oct. 2012, doi: 10.1016/S0034-4257(00)00119-X.

[10] K. M. Byrne, W. K. Lauenroth, P. B. Adler, and C. M. Byrne, "Estimating aboveground net primary production in grasslands: A comparison of nondestructive methods," *Rangeland Ecol. Manage.*, vol. 64, no. 5, pp. 498–505, Sep. 2011, doi: 10.2111/REM-D-10-00145.1.

[11] C. Eisfelder, C. Kuenzer, and S. Dech, "Derivation of biomass information for semi-arid areas using remote-sensing data," *Int. J. Remote Sens.*, vol. 33, no. 9, pp. 2937–2984, May 2012, doi: 10.1080/01431161.2011.620034.

[12] S. Moreau, R. Bosseno, X. F. Gu, and F. Baret, "Assessing the biomass dynamics of Andean *bofedal* and *totora* high-protein wetland grasses from NOAA/AVHRR," *Remote Sens. Environ.*, vol. 85, no. 4, pp. 516–529, Jun. 2003, doi: 10.1016/S0034-4257(00)00119-X.

[13] L. Jianlong, L. Tiangang, and C. Quangong, "Estimating grassland yields using remote sensing and GIS technologies in China," *New Zealand J. Agricult. Res.*, vol. 41, no. 1, pp. 31–38, Jan. 1998, doi: 10.1080/00288233.1998.9513285.

[14] J. Ge, "Remote sensing estimation and change analysis of grassland aboveground biomass in Northern China," Ph.D. thesis, Lanzhou Univ., Lanzhou, China, 2022, doi: 10.27204/d.cnki.glzhu.2022.000068.

[15] Y. Wang et al., "Prediction of aboveground grassland biomass on the Loess Plateau, China, using a random forest algorithm," *Sci. Rep.*, vol. 7, no. 1, p. 6940, Jul. 2017, doi: 10.1038/s41598-017-07197-6.

[16] C. Wu et al., "Comparison of machine-learning methods for above-ground biomass estimation based on Landsat imagery," *J. Appl. Remote Sens.*, vol. 10, no. 3, Aug. 2016, Art. no. 035010, doi: 10.1117/1.JRS.10.035010.

[17] H. Ren, B. Zhang, and X. Guo, "Estimation of litter mass in non-growing seasons in arid grasslands using MODIS satellite data," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 222–230, Jan. 2018, doi: 10.1080/22797254.2017.1418186.

[18] I. Borra-Serrano et al., "Canopy height measurements and non-destructive biomass estimation of *Loliumperenne* swards using UAV imagery," *Grass Forage Sci.*, vol. 74, no. 3, pp. 356–369, Jul. 2019, doi: 10.1111/gfs.12439.

[19] S. Idowu, S. Saguna, C. Åhlund, and O. Schelén, "Applied machine learning: Forecasting heat load in district heating system," *Energy Buildings*, vol. 133, pp. 478–488, Dec. 2016, doi: 10.1016/j.enbuild.2016.09.068.

[20] J. Wang et al., "Estimating leaf area index and aboveground biomass of grazing pastures using Sentinel-1, Sentinel-2 and Landsat images," *ISPRS J. Photogramm. Remote Sens.*, vol. 154, pp. 189–201, Aug. 2019, doi: 10.1016/j.isprsjprs.2019.06.007.

[21] G. P. Asner, "Biophysical and biochemical sources of variability in canopy reflectance," *Remote Sens. Environ.*, vol. 64, no. 3, pp. 234–253, Jun. 1998, doi: 10.1016/S0034-4257(98)00014-5.

[22] P. L. Nagler, C. S. T. Daughtry, and S. N. Goward, "Plant litter and soil reflectance," *Remote Sens. Environ.*, vol. 71, no. 2, pp. 207–215, Feb. 2000, doi: 10.1016/S0034-4257(99)00082-6.

[23] C. Ji, X. Li, H. Wei, and S. Li, "Comparison of different multispectral sensors for photosynthetic and non-photosynthetic vegetation-fraction retrieval," *Remote Sens.*, vol. 12, no. 1, p. 115, Jan. 2020, doi: 10.3390/rs12010115.

[24] K. Berger et al., "Assessing non-photosynthetic cropland biomass from spaceborne hyperspectral imagery," *Remote Sens.*, vol. 13, no. 22, p. 4711, Nov. 2021, doi: 10.3390/rs13224711.

[25] S. D. Blanchard, M. K. Jakubowski, and M. Kelly, "Object-based image analysis of downed logs in disturbed forested landscapes using LiDAR," *Remote Sens.*, vol. 3, no. 11, pp. 2420–2439, Nov. 2011, doi: 10.3390/rs3112420.

[26] X. Li, G. Zheng, J. Wang, C. Ji, B. Sun, and Z. Gao, "Comparison of methods for estimating fractional cover of photosynthetic and non-photosynthetic vegetation in the Otindag Sandy Land using GF-1 wide-field view data," *Remote Sens.*, vol. 8, no. 10, p. 800, Sep. 2016, doi: 10.3390/rs8100800.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:10:17 UTC from IEEE Xplore. Restrictions apply.

[27] C. I. Chang, *Hyperspectral Data Exploitation: Theory and Applications*, 1st ed. Baltimore, MD, USA: Wiley, 2007, pp. 1–18.

[28] X. Bai, W. Zhao, S. Ji, R. Qiao, C. Dong, and X. Chang, "Estimating fractional cover of non-photosynthetic vegetation for various grasslands based on CAI and DFI," *Ecol. Indicators*, vol. 131, Nov. 2021, Art. no. 108252, doi: 10.1016/j.ecolind.2021.108252.

[29] W. Song et al., "A remote sensing based forage biomass yield inversion model of alpine-cold meadow during grass-withering period in Sanjiangyuan area," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 17, Mar. 2014, Art. no. 012042, doi: 10.1088/1755-1315/17/1/012042.

[30] Z. Li and X. Guo, "Non-photosynthetic vegetation biomass estimation in semiarid Canadian mixed grasslands using ground hyperspectral data, Landsat 8 OLI, and Sentinel-2 images," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 6893–6913, Oct. 2018, doi: 10.1080/01431161.2018.1468105.

[31] G. Serbin, C. S. T. Daughtry, E. R. Hunt, J. B. Reeves, and D. J. Brown, "Effects of soil composition and mineralogy on remote sensing of crop residue cover," *Remote Sens. Environ.*, vol. 113, no. 1, pp. 224–238, Jan. 2009, doi: 10.1016/j.rse.2008.09.004.

[32] X. Cao, J. Chen, B. Matsushita, and H. Imura, "Developing a MODIS-based index to discriminate dead fuel from photosynthetic vegetation and soil background in the Asian steppe area," *Int. J. Remote Sens.*, vol. 31, no. 6, pp. 1589–1604, Mar. 2010, doi: 10.1080/01431160903475274.

[33] S. Huang, R. L. Crabtree, C. Potter, and P. Gross, "Estimating the quantity and quality of coarse woody debris in Yellowstone post-fire forest ecosystem from fusion of SAR and optical data," *Remote Sens. Environ.*, vol. 113, no. 9, pp. 1926–1938, Sep. 2009, doi: 10.1016/j.rse.2009.05.001.

[34] C. S. T. Daughtry, "Discriminating crop residues from soil by shortwave infrared reflectance," *Agronomy J.*, vol. 93, no. 1, pp. 125–131, Jan. 2001, doi: 10.2134/agronj2001.931125x.

[35] H. McNairn, C. Duguay, J. Boisvert, E. Huffman, and B. Brisco, "Defining the sensitivity of multi-frequency and multi-polarized radar backscatter to post-harvest crop residue," *Can. J. Remote Sens.*, vol. 27, no. 3, pp. 247–263, Jun. 2001, doi: 10.1080/07038992.2001.10854941.

[36] J. Delegido, J. Verrelst, J. P. Rivera, A. Ruiz-Verdú, and J. Moreno, "Brown and green LAI mapping through spectral indices," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 35, pp. 350–358, Mar. 2015, doi: 10.1016/j.jag.2014.10.001.

[37] S. S. Saatchi, J. J. van Zyl, and G. Asrar, "Estimation of canopy water content in Konza Prairie grasslands using synthetic aperture radar measurements during FIFE," *J. Geophys. Research: Atmos.*, vol. 100, no. D12, pp. 25481–25496, Dec. 1995, doi: 10.1029/95JD00852.

[38] B. Zheng, J. B. Campbell, G. Serbin, and J. M. Galbraith, "Remote sensing of crop residue and tillage practices: Present capabilities and future prospects," *Soil Tillage Res.*, vol. 138, pp. 26–34, May 2014, doi: 10.1016/j.still.2013.12.009.

[39] M. T. Eismann and R. C. Hardie, "Hyperspectral resolution enhancement using high-resolution multispectral imagery with arbitrary response functions," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 455–465, Mar. 2005, doi: 10.1109/TGRS.2004.837324.

[40] G. Qi, C. Chang, W. Yang, and G. Zhao, "Soil salinity inversion in coastal cotton growing areas: An integration method using satellite-ground spectral fusion and satellite-UAV collaboration," *Land Degradation Develop.*, vol. 33, no. 13, pp. 2289–2302, Aug. 2022, doi: 10.1002/ldr.4287.

[41] R. B. Gomez, A. Jazaeri, and M. Kafatos, "Wavelet-based hyperspectral and multispectral image fusion," *Proc. SPIE*, vol. 4383, pp. 36–42, Jun. 2001, doi: 10.1117/12.428249.

[42] H. Song, B. Huang, K. Zhang, and H. Zhang, "Spatio-spectral fusion of satellite images based on dictionary-pair learning," *Inf. Fusion*, vol. 18, pp. 148–160, Jul. 2014, doi: 10.1016/j.inffus.2013.08.005.

[43] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015, doi: 10.1109/TGRS.2014.2381272.

[44] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Bayesian fusion of hyperspectral and multispectral images," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Florence, Italy, May 2014, pp. 3176–3180, doi: 10.1109/ICASSP.2014.6854186.

[45] T. M. Tu, S. C. Su, H. C. Shyu, and P. S. Huang, "A new look at IHS-like image fusion methods," *Inf. Fusion*, vol. 2, pp. 177–186, May 2001, doi: 10.1016/S1566-2535(01)00036-7.

[46] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008, doi: 10.1109/TGRS.2008.916211.

[47] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012, doi: 10.1109/JSTARS.2012.2194696.

[48] H. Chen, Y. Ma, A. Zhu, Z. Wang, G. Zhao, and Y. Wei, "Soil salinity inversion based on differentiated fusion of satellite image and ground spectra," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 101, Sep. 2021, Art. no. 102360, doi: 10.1016/j.jag.2021.102360.

[49] L. Zhao, Y. Shi, B. Liu, C. Hovis, Y. Duan, and Z. Shi, "Finer classification of crops by fusing UAV images and Sentinel-2A data," *Remote Sens.*, vol. 11, no. 24, p. 3012, Dec. 2019, doi: 10.3390/rs11243012.

[50] D. D. Lee and H. S. Seung, "Learning the parts of objects by non-negative matrix factorization," *Nature*, vol. 401, no. 6755, pp. 788–791, Oct. 1999, doi: 10.1038/44565.

[51] B. Ren, L. Pueyo, G. B. Zhu, J. Debes, and G. Duchêne, "Non-negative matrix factorization: Robust extraction of extended structures," *Astrophysical J.*, vol. 852, no. 2, p. 104, Jan. 2018, doi: 10.3847/1538-4357/aaa1f2.

[52] J. Gan, T. Liu, L. Li, and J. Zhang, "Non-negative matrix factorization: A survey," *Comput. J.*, vol. 64, no. 7, pp. 1080–1092, Aug. 2021, doi: 10.1093/comjnl/bxab103.

[53] S. Madonsela, M. Cho, A. Ramoelo, and O. Mutanga, "Investigating the relationship between tree species diversity and Landsat-8 spectral heterogeneity across multiple phenological stages," *Remote Sens.*, vol. 13, no. 13, p. 2467, Jun. 2021, doi: 10.3390/rs13132467.

[54] F. Z. Benhalouche, M. S. Karoui, Y. Deville, and A. Ouamri, "Hyperspectral and multispectral data fusion based on linear-quadratic nonnegative matrix factorization," *J. Appl. Remote Sens.*, vol. 11, no. 2, May 2017, Art. no. 025008, doi: 10.1117/1.JRS.11.025008.

[55] F. Yang, F. Ma, S. Huo, and Y. Wang, "Multi-sensor data fusion of remotely-sensed images with sparse and logarithmic low-rank regularization for shadow removal and denoising," *Int. J. Remote Sens.*, vol. 42, no. 18, pp. 6961–6983, Sep. 2021, doi: 10.1080/01431161.2021.1941388.

[56] O. Berné, A. Helens, P. Pilleri, and C. Joblin, "Non-negative matrix factorization pansharpening of hyperspectral data: An application to mid-infrared astronomy," in *Proc. 2nd Workshop Hyperspectral Image Signal Process. Evol. Remote Sens.*, Reykjavik, Iceland, Jun. 2010, pp. 1–4, doi: 10.1109/WHISPERS.2010.5594900.

[57] Y. Liu, X.-Z. Pan, R.-J. Shi, Y.-L. Li, C.-K. Wang, and Z.-T. Li, "Predicting soil salt content over partially vegetated surfaces using non-negative matrix factorization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 11, pp. 5305–5316, Nov. 2015, doi: 10.1109/JSTARS.2015.2478490.

[58] J. Yang and X. Huang, "The 30 m annual land cover dataset and its dynamics in China from 1990 to 2019," *Earth Syst. Sci. Data*, vol. 13, no. 8, pp. 3907–3925, Aug. 2021, doi: 10.5194/essd-13-3907-2021.

[59] R. M. David, N. J. Rosser, and D. N. M. Donoghue, "Improving above ground biomass estimates of Southern Africa dryland forests by combining Sentinel-1 SAR and Sentinel-2 multispectral imagery," *Remote Sens. Environ.*, vol. 282, Dec. 2022, Art. no. 113232, doi: 10.1016/j.rse.2022.113232.

[60] R. A. Crabbe, D. W. Lamb, C. Edwards, K. Andersson, and D. Schneider, "A preliminary investigation of the potential of Sentinel-1 radar to estimate pasture biomass in a grazed, native pasture landscape," *Remote Sens.*, vol. 11, no. 7, p. 872, Apr. 2019, doi: 10.3390/rs11070872.

[61] M. Main-Knorn, B. Pflug, J. Louis, V. Debaecker, U. Müller-Wilm, and F. Gascon, "Sen2Cor for Sentinel-2," *Proc. SPIE*, vol. 10427, pp. 37–48, Oct. 2017.

[62] S. Cundill, H. van der Werff, and M. van der Meijde, "Adjusting spectral indices for spectral response function differences of very high spatial resolution sensors simulated from field spectra," *Sensors*, vol. 15, no. 3, pp. 6221–6240, Mar. 2015, doi: 10.3390/s150306221.

[63] H. McNairn and R. Protz, "Mapping corn residue cover on agricultural fields in Oxford County, Ontario, using Thematic Mapper," *Can. J. Remote Sens.*, vol. 19, no. 2, pp. 152–159, Apr. 1993, doi: 10.1080/07038992.1993.10874543.

[64] J. Qi et al., "RANGES improves satellite-based information and land cover assessments in Southwest United States," *Eos, Trans. Amer. Geophys. Union*, vol. 83, no. 51, pp. 601–606, Dec. 2002, doi: 10.1029/2002EO000411.

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[65] M. A. Hardisky, R. M. Smart, and V. Klemas, "Seasonal spectral characteristics and aboveground biomass of the tidal marsh plant," *Spartina–Alterniflora Photogramm. Eng. Remote Sens.*, vol. 49, no. 1, pp. 85–92, Jan. 1983.

[66] L. Vescovo and D. Gianelle, "Using the MIR bands in vegetation indices for the estimation of grassland biophysical parameters from satellite remote sensing in the Alps region of Trentino (Italy)," *Adv. Space Res.*, vol. 41, no. 11, pp. 1764–1772, Jan. 2008, doi: 10.1016/j.asr.2007.07.043.

[67] A. P. van Deventer, A. D. Ward, P. H. Gowda, and J. G. Lyon, "Using Thematic Mapper data to identify contrasting soil plains and tillage practices," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 1, pp. 87–93, 1997.

[68] T. Motohka, K. N. Nasahara, H. Oguma, and S. Tsuchida, "Applicability of green-red vegetation index for remote sensing of vegetation phenology," *Remote Sens.*, vol. 2, no. 10, pp. 2369–2387, Oct. 2010, doi: 10.3390/rs2102369.

[69] A. A. Gitelson and M. N. Merzlyak, "Signature analysis of leaf reflectance spectra: Algorithm development for remote sensing of chlorophyll," *J. Plant Physiol.*, vol. 148, nos. 3–4, pp. 494–500, 1996, doi: 10.1016/S0176-1617(96)80284-7.

[70] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, May 1979, doi: 10.1016/0034-4257(79)90013-0.

[71] J. W. Rouse, R. H. Haas, J. A. Schell, and D. W. Deering, "Monitoring vegetation systems in the Great Plains with ERTS," in *Proc. NASA Goddard Space Flight Center 3D ERTS-1 Symp.*, vol. 1, Jan. 1974, p. 309.

[72] B. Datt, "A new reflectance index for remote sensing of chlorophyll content in higher plants: Tests using eucalyptus leaves," *J. Plant Physiol.*, vol. 154, no. 1, pp. 30–36, Jan. 1999, doi: 10.1016/S0176-1617(99)80314-9.

[73] D. A. Sims and J. A. Gamon, "Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages," *Remote Sens. Environ.*, vol. 81, nos. 2–3, pp. 337–354, Aug. 2002.

[74] M. Zhang, B. Wu, and J. Meng, "Quantifying winter wheat residue biomass with a spectral angle index derived from China Environmental Satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 32, pp. 105–113, Oct. 2014, doi: 10.1016/j.jag.2014.03.020.

[75] D. Poobathy and R. M. Chezian, "Edge detection operators: Peak signal to noise ratio based comparison," *Int. J. Image, Graph. Signal Process.*, vol. 6, no. 10, pp. 55–61, Sep. 2014, doi: 10.5815/ijigsp.2014.10.07.

[76] L. Li and Y. J. Zhang, "A survey of algorithms of non-negative matrix factorization," *Acta Electron. Sinica*, vol. 36, no. 4, pp. 737–743. Apr. 2008.

[77] A. Horé and D. Ziou, "Is there a relationship between peak-signal-to-noise ratio and structural similarity index measure?" *IET Image Process.*, vol. 7, no. 1, pp. 12–24, Feb. 2013, doi: 10.1049/iet-ipr.2012.0489.

[78] R. Badeau, N. Bertin, and E. Vincent, "Stability analysis of multiplicative update algorithms for non-negative matrix factorization," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Prague, Czech Republic, May 2011, pp. 2148–2151, doi: 10.1109/ICASSP.2011.5946752.

[79] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. Roy. Stat. Soc. B, Methodol.*, vol. 58, no. 1, pp. 267–288, Jan. 1996.

[80] A. Dallakyan, "Graphiclasso: Graphical lasso for learning sparse inverse-covariance matrices," *Stata J., Promoting Commun. Statist. Stata*, vol. 22, no. 3, pp. 625–642, Sep. 2022, doi: 10.1177/1536867X221124538.

[81] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, Oct. 2011, doi: 10.1023/A:1010933404324.

[82] S. M. Ghosh, M. D. Behera, B. Jagadish, A. K. Das, and D. R. Mishra, "A novel approach for estimation of aboveground biomass of a carbon-rich mangrove site in India," *J. Environ. Manage.*, vol. 292, Aug. 2021, Art. no. 112816, doi: 10.1016/j.jenvman.2021.112816.

[83] Y. Tian et al., "Aboveground mangrove biomass estimation in Beibu Gulf using machine learning and UAV remote sensing," *Sci. Total Environ.*, vol. 781, Aug. 2021, Art. no. 146816, doi: 10.1016/j.scitotenv.2021.146816.

[84] N. Zeng, X. Ren, H. He, L. Zhang, P. Li, and Z. Niu, "Estimating the grassland aboveground biomass in the three-river headwater region of China using machine learning and Bayesian model averaging," *Environ. Res. Lett.*, vol. 16, no. 11, Nov. 2021, Art. no. 114020, doi: 10.1088/1748-9326/ac2e85.

[85] C. Willmott and K. Matsuura, "Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance," *Climate Res.*, vol. 30, pp. 79–82, Dec. 2005, doi: 10.3354/cr030079.

[86] C. S. T. Daughtry, J. E. Mcmurtrey, E. W. Chappelle, W. J. Hunter, and J. L. Steiner, "Measuring crop residue cover using remote sensing techniques," *Theor. Appl. Climatol.*, vol. 54, nos. 1–2, pp. 17–26, Mar. 1996, doi: 10.1007/BF00863555.

[87] Y. Sun, Q. Qin, H. Ren, T. Zhang, and S. Chen, "Red-edge band vegetation indices for leaf area index estimation from Sentinel-2/MSI imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 826–840, Feb. 2020, doi: 10.1109/TGRS.2019.2940826.

[88] X. H. Zhang et al., "Estimating aboveground biomass of grassland in Zoige by visible vegetation index derived from unmanned aerial vehicle image," *Remote Sens. Technol. Appl.*, vol. 31, no. 1, p. 51, 2016. [Online]. Available: http://www.rsta.ac.cn/EN/10.11873/j.issn.1004-0323.2016.1.0051

[89] B. Du, S. Wang, N. Wang, L. Zhang, D. Tao, and L. Zhang, "Hyperspectral signal unmixing based on constrained non-negative matrix factorization approach," *Neurocomputing*, vol. 204, pp. 153–161, Sep. 2016, doi: 10.1016/j.neucom.2015.10.132.

[90] N. Yokoya, T. Yairi, and A. Iwasaki, "Hyperspectral, multispectral, and panchromatic data fusion based on coupled non-negative matrix factorization," in *Proc. 3rd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Lisbon, Portugal, Jun. 2011, pp. 1–4, doi: 10.1109/WHISPERS.2011.6080924.

[91] J. Li, G. Zhou, Y. Qiu, Y. Wang, Y. Zhang, and S. Xie, "Deep graph regularized non-negative matrix factorization for multi-view clustering," *Neurocomputing*, vol. 390, pp. 108–116, May 2020, doi: 10.1016/j.neucom.2019.12.054.

[92] R. W. Tuan, "Research on hyperspectral and multispectral image fusion," Ph.D. thesis, Hunan Univ., Hunan, China, 2020, doi: 10.27135/d.cnki.ghudu.2020.002186.

[93] B. Hu, Q. Li, and A. Smith, "Noise reduction of hyperspectral data using singular spectral analysis," *Int. J. Remote Sens.*, vol. 30, no. 9, pp. 2277–2296, May 2009, doi: 10.1080/01431160802549344.

[94] A. Horé and D. Ziou, "Image quality metrics: PSNR vs. SSIM," in *Proc. 20th Int. Conf. Pattern Recognit.*, Istanbul, Turkey, Aug. 2010, pp. 2366–2369, doi: 10.1109/ICPR.2010.579.

[95] Y. Kawakami and M. Sugiyama, "Investigating overparameterization for non-negative matrix factorization in collaborative filtering," in *Proc. 15th ACM Conf. Recommender Syst.* Amsterdam, The Netherlands: ACM, Sep. 2021, pp. 685–690, doi: 10.1145/3460231.3478854.

[96] Z. Li and X. Guo, "Can polarimetric Radarsat-2 images provide a solution to quantify non-photosynthetic vegetation biomass in semiarid mixed grassland?" *Can. J. Remote Sens.*, vol. 43, no. 6, pp. 593–607, Nov. 2017, doi: 10.1080/07038992.2017.1394181.

[97] H. McNairn, C. Duguay, B. Brisco, and T. J. Pultz, "The effect of soil and crop residue characteristics on polarimetric radar response," *Remote Sens. Environ.*, vol. 80, no. 2, pp. 308–320, May 2002.

[98] N. Baghdadi, M. Zribi, C. Loumagne, P. Ansart, and T. Anguela, "Analysis of TerraSAR-X data and their sensitivity to soil surface parameters over bare agricultural fields," *Remote Sens. Environ.*, vol. 112, no. 12, pp. 4370–4379, Dec. 2008, doi: 10.1016/j.rse.2008.08.004.

[99] C. H. Q. Ding, T. Li, and M. I. Jordan, "Convex and semi-nonnegative matrix factorizations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 1, pp. 45–55, Jan. 2010, doi: 10.1109/TPAMI.2008.277.

[100] X. Liu et al., "Comprehensive growth index monitoring of desert steppe grassland vegetation based on UAV hyperspectral," *Frontiers Plant Sci.*, vol. 13, Jan. 2023, Art. no. 1050999, doi: 10.3389/fpls.2022.1050999.

[101] P. Hu, W. Guo, S. C. Chapman, Y. Guo, and B. Zheng, "Pixel size of aerial imagery constrains the applications of unmanned aerial vehicle in crop breeding," *ISPRS J. Photogramm. Remote Sens.*, vol. 154, pp. 1–9, Aug. 2019, doi: 10.1016/j.isprsjprs.2019.05.008.

[102] K. Anderson and K. J. Gaston, "Lightweight unmanned aerial vehicles will revolutionize spatial ecology," *Frontiers Ecol. Environ.*, vol. 11, no. 3, pp. 138–146, Apr. 2013, doi: 10.1890/120150.

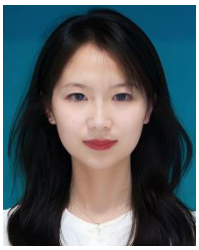
[103] S. L. Ustin and E. M. Middleton, "Current and near-term advances in Earth observation for ecological applications," *Ecol. Processes*, vol. 10, no. 1, Jan. 2021, doi: 10.1186/s13717-020-00255-4.

[104] J. Verrelst et al., "Prototyping vegetation traits models in the context of the hyperspectral chime mission preparation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Brussels, Belgium, Jul. 2021, pp. 7678–7681, doi: 10.1109/IGARSS47720.2021.9554407.

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[105] I. Numata, D. Roberts, O. Chadwick, J. Schimel, L. Galvao, and J. Soares, "Evaluation of hyperspectral data for pasture estimate in the Brazilian Amazon using field and imaging spectrometers," *Remote Sens. Environ.*, vol. 112, no. 4, pp. 1569–1583, Apr. 2008, doi: 10.1016/j.rse.2007.08.014.

[106] A. Bannari, A. Pacheco, K. Staenz, H. McNairn, and K. Omari, "Estimating and mapping crop residues cover on agricultural lands using hyperspectral and IKONOS data," *Remote Sens. Environ.*, vol. 104, no. 4, pp. 447–459, Oct. 2006, doi: 10.1016/j.rse.2006.05.018.



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